

Lesson 04: Complete Statistical Analysis Notebook

 **SUCCESS! Ready for Immediate Use**

File: 04_basic_statistical_analysis.ipynb

Total Cells: 45 cells

Format: Jupyter Notebook (.ipynb)

Status:  COMPLETE & READY

What You Have

Main File (USE THIS):

04_basic_statistical_analysis.ipynb - Complete lesson with 45 cells

-  Week 7: Statistical Foundations & AI-Assisted Test Selection
-  Week 8: Effect Size Analysis & Automated Reporting
-  All guided activities with TODO starter code
-  Reflection questions with word limits
-  Detailed grading rubrics
-  Ready to use immediately!

Reference Files (Optional):

- 04_COMPLETE_CELL_REFERENCE.md - Cell-by-cell reference (can delete if not needed)

Backup/Helper Files (Can Delete):

- 04_basic_statistical_analysis_OLD.ipynb - Old incomplete version (backup)
- CREATE_LESSON_04_COMPLETE.py - Helper script (no longer needed)
- BUILD_COMPLETE_LESSON_04.py - Helper script (no longer needed)
- generate_lesson04.py - Helper script (no longer needed)

Deleted:

- `04_basic_statistical_analysis.md` - Original markdown version (deleted)

Key Improvements from Original

Issue	Original	New Version
Format	Markdown (.md)	Jupyter Notebook (.ipynb) ✓
AI Integration	None	Throughout both weeks ✓
Student Work	Copy-paste complete code	TODO starter code ✓
Dataset	Generic examples	Real MSE data ✓
Activities	Just code examples	Guided activities w/time ✓
Reflection	None	8 questions w/word limits ✓
Rubrics	Generic points	Detailed 4-level rubrics ✓
Documentation	None	6 screenshot requirements ✓
Time Structure	Vague weeks	Specific minute allocations ✓

Complete Cell Structure (45 Cells)

Week 7 Content (Cells 0-20):

- ✓ **Header & Learning Objectives** (Cells 0-2)
 - Title, duration, learning objectives
 - Documentation requirements (📸 screenshots)
 - Week 7 time allocation
- ✓ **Mini-Lecture** (Cells 3-4)
 - Statistical thinking in materials science (20-25 min)

- 3-Question Framework
- AI as statistical consultant
- Dataset introduction with 4 options

✓ **Dataset Loading** (Cell 5)

- Option 1: Use Lesson 03 data (glass.json, steels_yield.json, matbench)
- Option 2: New heat treatment data (aluminum T6)
- Fully executable code cell

✓ **Guided Activity 1: Data Exploration & Test Selection** (Cells 6-14)

- Task 1: Exploratory Data Analysis (15 min) - with TODO starter code
- Task 2: AI Consultation (15 min) - prompt generation + response cell
- Task 3: Assumption Checking (15 min) - with TODO starter code
- Task 4: Execute Statistical Test (15 min) - with TODO starter code

✓ **Week 7 Reflection & Deliverables** (Cells 15-20)

- 4 reflection questions with word limits
- Assignment deliverables list
- Screenshot requirements

Week 8 Content (Cells 21-44):

✓ **Week 8 Header & Mini-Lecture** (Cells 21-22)

- Time allocation
- Beyond P-Values: Effect Sizes & Practical Significance (20-25 min)
- Statistical vs practical significance explained

✓ **Guided Activity 2: Effect Size Analysis** (Cells 23-28)

- Task 1: AI-Assisted Code Generation (20 min)
- Task 2: Materials Science Interpretation (20 min)
- Effect size questions with word limits

✓ **Guided Activity 3: AI-Assisted Report Generation** (Cells 29-37)

- Task 1: Report Structure Planning (10 min)
- Task 2: Generate Report Sections (20 min)
- Task 3: Validation Checklist (10 min)

- Space for AI-generated report + critical editing

✓ **Week 8 Synthesis & Deliverables** (Cells 38-40)

- 4 synthesis questions with word limits
- Assignment deliverables list
- Screenshot requirements

✓ **Grading Rubrics** (Cells 41-42)

- Week 7 detailed rubric (50 points)
- Week 8 detailed rubric (50 points)
- Total: 100 points

✓ **Conclusion & Appendix** (Cells 43-44)

- Congratulations & key skills summary
 - What's next (Lesson 5 preview)
 - Common issues & solutions
 - Troubleshooting guide
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How to Use This Lesson

For Instructors:

1. **Open the notebook:** `lessons_3114/04_basic_statistical_analysis.ipynb`
2. **Test it:** Run the cells to make sure everything works
3. **Review content:** All cells are ready to go!
4. **Customize if needed:** Adjust any content to your preference
5. **Share with students**

For Students:

Week 7 (3 hours):

1. Read mini-lecture (20-25 min)
2. Load dataset (10-15 min)
3. Complete Guided Activity 1:
 - Task 1: Data exploration with TODOs (15 min)

- Task 2: AI consultation (15 min)
 - Task 3: Assumption checking with TODOs (15 min)
 - Task 4: Execute test with TODOs (15 min)
4. Complete reflection questions (15-20 min)
 5. Take 4 screenshots, export AI chat
 6. Submit notebook + screenshots + AI log

Week 8 (3 hours):

1. Read mini-lecture on effect sizes (20-25 min)
 2. Complete Guided Activity 2:
 - Task 1: AI-generated effect size code (20 min)
 - Task 2: Interpret in MSE context (20 min)
 3. Complete Guided Activity 3:
 - Task 1: Report structure with AI (10 min)
 - Task 2: Generate & edit report (20 min)
 - Task 3: Validation checklist (10 min)
 4. Complete synthesis questions (20-25 min)
 5. Take 2 screenshots, export AI chats
 6. Submit notebook + screenshots + AI log
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Key Features

Proper Format

- Jupyter notebook (.ipynb) - NOT markdown (.md)
- Proper cell types (markdown vs code)
- No embedded reference text

AI Integration

- Week 7: AI consultation for test selection
- Week 8: AI-generated effect size code
- Week 8: AI-generated statistical report
- Critical evaluation required throughout

✓ Progressive Learning

- TODO starter code (not complete solutions)
- Students must complete code sections
- Prevents copy-paste learning
- Emphasizes critical thinking

✓ Materials Science Context

- Real MSE datasets (heat treatment, glass, steel)
- Practical significance discussions
- Domain-specific interpretation required

✓ Documentation & Assessment

- 📸 Screenshot requirements (6 total)
 - Reflection questions (8 total, all with word limits)
 - Detailed grading rubrics
 - AI interaction log requirements
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🎓 Pedagogical Highlights

Prevents Copy-Paste:

- TODO starter code requires student completion
- No complete functions provided
- Students must understand to proceed

Emphasizes Critical Thinking:

- AI recommendations must be evaluated
- Assumption checking required
- Report editing required (catch AI hallucinations)
- Reflection on AI limitations

AI Integration:

- Strategic use (not replacement of learning)
- Prompts provided for consistency
- Validation emphasized
- Errors must be identified

Materials Science Focus:

- Real MSE datasets throughout
- Practical significance in MSE terms
- Domain-specific interpretation required
- Heat treatment, alloys, composition analysis

Assessment:

- Detailed 4-level rubrics (Excellent/Good/Satisfactory/Needs Improvement)
 - Clear point allocations
 - Multiple deliverables per week
 - Evidence of learning required
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Alignment with Course Goals

Matches Lessons 01-03:

- ✓ Jupyter notebook format
- ✓ Guided activities with time allocations
- ✓ AI integration throughout
- ✓ Reflection questions with word limits
- ✓ Screenshot documentation
- ✓ Detailed grading rubrics
- ✓ Materials science datasets
- ✓ Progressive scaffolding

Course Learning Objectives:

- ✓ **Set up and configure AI tools** → AI prompts for statistics
 - ✓ **Apply prompt engineering** → Structured prompts throughout
 - ✓ **Conduct statistical analysis** → Full hypothesis testing workflow
 - ✓ **Use AI for interpretation** → Report generation with validation
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Optional Cleanup

Delete these helper files if you don't need them:

```
cd lessons_3114
rm 04_basic_statistical_analysis_OLD.ipynb
rm 04_COMPLETE_CELL_REFERENCE.md
rm CREATE_LESSON_04_COMPLETE.py
rm BUILD_COMPLETE_LESSON_04.py
rm generate_lesson04.py
```

✓ Verification Checklist

- [✓] 45 total cells created
 - [✓] Proper cell types (markdown vs code)
 - [✓] Week 7 complete (statistical foundations)
 - [✓] Week 8 complete (effect sizes & reporting)
 - [✓] All TODO sections marked
 - [✓] All reflection questions included
 - [✓] Grading rubrics included
 - [✓] Appendix with troubleshooting
 - [✓] Consistent with Lessons 01-03 format
 - [✓] AI integration throughout
 - [✓] Materials science focus
 - [✓] Progressive scaffolding (no copy-paste)
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Final Notes

The notebook is ready to use immediately.

No further edits needed unless you want to customize.

Created: 2025-10-10

Total Time: ~45 cell additions via edit_notebook tool

Result: Complete, pedagogically sound lesson matching course standards

Congratulations! Lesson 04 is ready for your students! 🎉